Steering Large Language Models for Emotion Control: A Systematic Study of Prompts, Vectors, and Conceptors

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Abstract

Large Language Models (LLMs) excel at factual and semantic reasoning but struggle to convey and sustain emotional states in dialogue. This limitation hinders their use in empathetic interaction, education, and affective computing. Prior approaches to emotion-aware text generation rely on prompt engineering or fine-tuning, which are brittle and resource-intensive. Recent advances in activation steering enable lightweight, representation-level control, yet most existing work targets narrow attributes (e.g., single word translation, antonymy, capitalization) with limited evaluation in open-ended emotional response generation.

In this work, we systematically analyze emotional steering across two method families: (1) additive emotion vectors (EVs) extracted and, and (2) conceptor-based steering. Our study reveals that while additive EVs provide strong control over emotional aware outputs, vanilla conceptors fail to generalize due to activation bias and over-regularization for open ended tasks. Building on this diagnosis, we introduce CoRAL (Conceptor Residual Alignment Layer), a residual–affine reformulation that stabilizes conceptor steering without altering model semantics. Extensive ablations over layers, aperture scaling, and activation geometry (t-SNE) offer insight into controllability and failure modes. Together, these results establish a principled framework for analyzing and improving emotion aware steering in LLMs.

1 Introduction

Large Language Models (LLMs) achieve impressive results in reasoning and factual benchmarks, yet they often fail to maintain consistent emotional expression across text generation tasks. Outputs tend to drift toward neutral or generic tones, limiting their applicability in empathy-sensitive settings such as tutoring, counseling, and assistive communication (Bai et al., 2022). This challenge

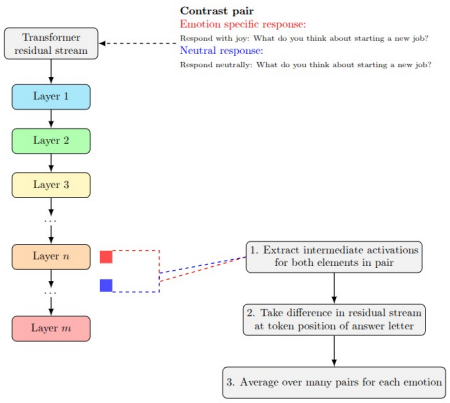


Figure 1: Steering Vector Generation

underscores the need for mechanisms that can modulate emotional style controllably and interpretably, without costly retraining.

Traditional approaches like prompt engineering and supervised fine-tuning only partially address this need. Prompts are brittle and highly context-dependent, while fine-tuning incurs high annotation and compute cost, scales poorly across emotional categories, and often harms generalization. To overcome these limitations, recent work explores activation-level steering, which manipulates a model's internal representations at inference time to guide behavior (Turner et al., 2023). Activation steering provides lightweight, model-agnostic control and forms the theoretical foundation of our study on emotion steering.

Within this paradigm, we focus on two main methods: (1) Additive Emotion Vectors (EVs)–mean-difference vectors extracted from emotionally tagged activations (Dong et al., 2025)–enable direct, linear modulation of affective style. We conduct extensive ablation studies on EVs, analyzing layer sensitivity, magnitude scaling, and geometric clustering (via t-SNE and spectral analysis) to understand how emotional subspaces form and interact. (2) Conceptor-based steering, introduced by

Jaeger (2014) and later adapted for transformers (Jørgensen et al., 2024), generalizes linear addition through covariance-aware "soft projections." Conceptors define ellipsoidal regions in activation space, allowing structured control and Boolean composition, but exhibit instability in emotional domains due to strong mean shifts and high intra-class variance.

Our analysis of these phenomena leads to CoRAL (Conceptor Residual Alignment Layer)–a residual–affine extension of conceptor steering that recenters activations and stabilizes projection while retaining semantic coherence. CoRAL unifies additive and conceptor-based approaches under a shared framework and supports interpretable, controllable emotional modulation in LLMs.

2 Background and Related Work

In this section, we review prior work on emotion-aware language generation, activation-based control, and conceptor steering, highlighting the strengths and limitations that motivate our proposed approach.

2.1 Emotion-Aware Text Generation

Controlling emotional expression in LLMs has been studied primarily through prompting and supervised fine-tuning. Controllable Emotion Generation by Dong et al. (2025) introduced the EmotionQuery and EmotionQuery+ benchmarks, enabling evaluation of discrete emotions using 100 prompt–response pairs per category. While effective for categorical affect control, these datasets focus on accuracy of discrete emotions rather than compositional or continuous affect spectra.

Madani et al. (2024) extended this idea to long emotional-support dialogues via fine-tuning LLaMA 2/3 models and proposed the Strategy-Relevant Attention metric to evaluate emotional consistency over multiple turns. However, their setup remains domain-specific and does not explore latent-space control or representation-level steering.

Similarly, Wang et al. (2024a) modeled affect as a stylistic attribute using style vectors, steering generative models toward sentiment or formality. Though conceptually adjacent, style vectors operate at a coarser granularity and cannot represent fine-grained emotional intensity or mixed emotions (e.g., frustration = sadness + anger).

To overcome the limited sample diversity of EmotionQuery+, we construct a synthetic extension, EmotionScape-Mini, using GPT-4o-mini with the same prompt template and parameters as Dong et al. (2025). This augmentation provides denser emotion coverage and supports our large-scale experiments on conceptor-based methods.

2.2 Activation Steering and Latent Interventions

A growing body of work manipulates hidden activations rather than model weights to control behavior at inference time.

Zhang et al. (2024) derive contrastive activation vectors to improve instruction following, demonstrating compositional control across constraints.

SADI (Wang et al., 2024b) introduces semantics-adaptive steering, adjusting vectors based on input context to reduce side effects from static steering directions.

Jorgensen and Albanie (2023) propose mean-centering of activations to mitigate global bias drift, improving stability in toxicity and genre control.

Despite these advances, most activation-steering methods remain purely additive, lacking mechanisms to adapt to covariance structure or residual alignment. Few studies examine how layer depth, steering magnitude, or representation geometry affect emotion control.

In this work, we extend additive Emotion Vectors (EVs) beyond fixed-vector operations by performing systematic ablation studies–analyzing layer sensitivity, scaling effects (λ), and latent-space geometry via t-SNE and spectral decomposition–to better understand the controllability and saturation of additive steering in emotional domains.

2.3 Conceptor Steering and Boolean Composition

Conceptor theory, introduced by Jaeger (2014), defines a soft projection operator derived from the covariance of internal states, allowing networks to recall and suppress patterns by projecting activations onto concept-specific subspaces.

The approach was later adapted to transformer activations by Jørgensen et al. (2024) in "Steering Large Language Models with Activation Deltas", where conceptors generalized linear steering vectors through affine projections and Boolean composition (AND/OR/NOT).

However, prior work on conceptor steering has only evaluated structured function tasks such as

antonymy, capitalization, translation, and plurality–tasks with clearly separable representations and deterministic outputs. No prior study has applied conceptor steering to behavioral, stylistic, or emotional open-ended generation, where activations exhibit strong mean shifts and high intra-class variance. As a result, conceptor steering's stability, interpretability, and controllability in affective domains remain untested.

We identify three key limitations of existing conceptor approaches:

- they assume zero-mean activations, which is invalid for emotion features;
- they ignore residual biases introduced by activation anisotropy; and
- they lack a mechanism to balance projection strength against semantic preservation.

2.4 Positioning of Our Work

Our work bridges the literature on emotion control (Dong et al., 2025) and activation-level steering (Zhang et al., 2024; Jorgensen and Albanie, 2023) with the theoretical foundation of conceptor projection (Jaeger, 2014; Jørgensen et al., 2024).

We show that classical conceptor steering collapses on affective tasks due to mean-shifted activations and propose CoRAL (Conceptor Residual Alignment Layer)–a residual–affine reformulation that recenters and stabilizes projection. CoRAL unifies additive Emotion Vectors and conceptor matrices within one framework, enabling continuous, interpretable control of emotional intensity.

Together with comprehensive ablations and geometric analyses, our study provides a principled methodological framework for diagnosing and improving emotion steering in large language models.

3 Proposed Methodology

Our goal is to develop structured and interpretable mechanisms for emotional steering in large language models (LLMs) through representation-space control. We focus on two complementary families of activation-level interventions: (1) Additive Emotion Vectors (EVs), representing first-order mean-shift steering, and (2) Conceptor-based steering, which introduces second-order, covariance-aware control. We further propose a new variant, CoRAL (Conceptor Residual Alignment Layer), that unifies both approaches under a single affine–residual formulation. Implementation details (datasets, models, and hyperparameters) are presented in section4.

3.1 Notation and Preliminaries

Let $h_\ell(x) \in \mathbb{R}^d$ denote the hidden activation of a model at layer ℓ for input x. We use the following notation consistently throughout:

Symbol	Definition
$\mathcal{E}_e, \mathcal{E}_0$	Emotion-conditioned and neutral prompt sets
$\mu_\ell^{(e)} = \mathbb{E}_{x\sim\mathcal{E}_e}[h_\ell(x)]$	Mean activation for emotion e
$\Sigma_\ell^{(e)} = \mathrm{Cov}_{x\sim\mathcal{E}_e}[h_\ell(x)]$	Covariance of activations
$v_\ell^{(e)} = \mu_\ell^{(e)} - \mu_\ell^{(0)}$	Additive emotion vector
$C_\ell^{(e)}$	Conceptor matrix at layer ℓ
α, γ	Aperture (regularization) parameters
λ	Steering strength at inference time
$b_\ell^{(e)}$	Bias term for mean correction in CoRAL

All activations and transformations are applied in $\mathbb{R}^d$. Throughout, expectation and covariance are computed over pooled token-level representations from the target layer.

3.2 Additive Emotion Vector (EV) Steering

3.2.1 Motivation

Additive steering has emerged as a simple yet powerful technique for controllable generation. Prior work showed that latent steering vectors, computed as differences in hidden activations between contrasting contexts, can be injected at inference to bias outputs toward desired behaviors such as safety, style, or emotion. Compared to prompt engineering, additive steering is less brittle and allows finer control at the representation level, but its performance depends critically on how the vectors are extracted and applied.

We adopt and extend this framework for emotional control, evaluating both baseline extraction methods and novel compositional variants.

3.2.2 Extraction of Emotion Vectors

We consider two extraction pipelines for an emotion e, both based on contrasting emotion-primed and neutral contexts:

Following Dong et al. (2025), we select a set of n neutral prompts $\{c_i\}_{i=1}^n$. For each, we collect hidden activations $h_\ell(c_i; e)$ when the model is prompted with emotion e, and $h_\ell(c_i; \text{neutral})$ for a

neutral variant. At layer ℓ, the emotion vector is:

$$v_e^{\text{prompt}} = \frac{1}{n}\sum_{i=1}^{n}\left(h_\ell(c_i; e) - h_\ell(c_i; \text{neutral})\right). \tag{1}$$

$$v_\ell^{\text{base}} = \frac{1}{|E|}\sum_{e\in E} v_e^{\text{ft}}, \tag{2}$$

v_ℓ^{base} denotes average EV vectora for all emotions and is used to get emotional aware outputs as compared to emotion specific outputs,denoted by v_e^{prompt}. This construction captures average shifts in representation space induced by the target emotion.

3.2.3 Injection of Emotion Vectors

At inference, the EV is added to hidden activations:

$$h'_\ell = h_\ell + \lambda \cdot v_e, \tag{3}$$

where λ is a tunable scalar controlling steering strength.

We test:

- **Single-layer injection:** adding at one fixed layer.
- **Multi-layer injection:** adding the same vector at multiple layers.
- **Scaling strategies:** following Anthropic's contrastive activation addition, we normalize vectors layer-wise and adjust magnitudes to prevent over-steering.



Figure 2: Emotion vector steering response examples

3.3 Conceptor Steering (Original Formulation)

Conceptor theory, introduced by Jaeger (2014), provides a principled framework for representing and manipulating subspaces of neural activations. A conceptor can be interpreted as a soft projection operator that captures the correlation structure of a specific activation manifold–allowing selective filtering and controlled modulation of latent representations. Conceptually, it behaves like a tunable linear filter that passes directions characteristic of a target concept while attenuating orthogonal components.

For an emotion-conditioned activation distribution $\mathcal{E}_e$ at layer ℓ, with empirical covariance $\Sigma_\ell^{(e)} = \text{Cov}_{x\sim\mathcal{E}_e}[h_\ell(x)]$, the corresponding conceptor matrix is defined as

$$C_\ell^{(e)} = \Sigma_\ell^{(e)}\left(\Sigma_\ell^{(e)} + \alpha^{-2} I\right)^{-1}, \tag{4}$$

where $\alpha > 0$ denotes the aperture or regularization parameter controlling the selectivity of projection. Smaller aperture values make the conceptor more restrictive, while larger values allow broader reconstruction of the original activation space.

Given this formulation, the standard conceptor projection rule modifies activations as

$$h'_\ell(x) = C_\ell^{(e)} h_\ell(x), \tag{5}$$

thereby enforcing a soft alignment between the current hidden state and the manifold associated with the target emotion e. When applied to language models, this operation can be viewed as steering the model's internal representation toward the subspace spanned by emotion-specific activations.

However, direct application of this formulation exhibits several drawbacks when used for emotion steering or open-ended text control:

Violation of zero-mean assumption. The conceptor derivation assumes that activations are zero-centered. Emotional activations, however, often show significant mean shifts ($\mu_\ell^{(e)} \neq 0$), so applying $C_\ell^{(e)}$ directly tends to suppress informative mean components that encode emotional tone.

Loss of expressive dimensions. Since $C_\ell^{(e)}$ acts as a low-pass filter in the covariance domain, it attenuates high-variance directions that correspond to stylistic or affective expressivity. The resulting generations tend to become neutral or monotonous in affect.

Over-regularization under high covariance noise. In transformer representations with limited samples per emotion set, $\Sigma_\ell^{(e)}$ may be ill-conditioned. The resulting projection $h' = Ch$ then behaves like an overly smoothed estimate, collapsing informative subspace variance. Empirically, this yields degenerate or incoherent generations, as similarly observed by Jørgensen et al. (2024).

In summary, while the original conceptor formulation offers an elegant linear control mechanism, its assumptions and filtering behavior lead to loss of semantic richness when applied to emotional or stylistic steering. These limitations motivate the bias-corrected and residual extension described next.

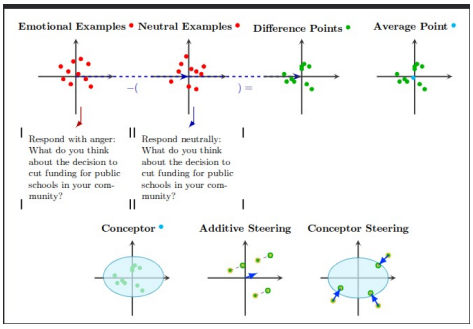


Figure 3: Comparing Conceptor based and additive steering

3.4 CoRAL: Conceptor Residual Alignment Layer

While the original conceptor formulation introduced by Jaeger (2014) provides an elegant linear mechanism for subspace control, its zero-mean and purely multiplicative nature often destabilizes emotion-conditioned activations in large language models. As also noted by Jørgensen et al. (2024), the mean shift of hidden representations carries crucial affective information; removing it through naïve conceptor projection yields semantically degraded, affect-neutral text.

To overcome these limitations, we propose the Conceptor Residual Alignment Layer (CoRAL)–a bias-corrected, residual extension that preserves semantic context while enabling controlled alignment toward emotion-specific manifolds.

3.4.1 Motivation

The standard projection $h'_\ell = C_\ell^{(e)} h_\ell$ implicitly assumes zero-mean activations. In practice, emotion-conditioned representations are mean-shifted ($\mu_\ell^{(e)} \neq 0$), so direct projection acts as a zero-mean low-pass filter that collapses expressive directions and produces incoherent or neutralized text.

CoRAL addresses this by introducing (1) mean correction to retain affective centroids, and (2) a residual formulation that softly adjusts rather than replaces activations, thus maintaining contextual semantics.

3.4.2 Derivation

Starting with the standard conceptor transformation,

$$h'_\ell = C_\ell^{(e)} h_\ell, \tag{6}$$

the expected mean becomes

$$\mathbb{E}[h'_\ell] = C_\ell^{(e)} \mu_\ell^{(e)} \neq \mu_\ell^{(e)}, \tag{7}$$

revealing a systematic mean distortion.

We therefore introduce a bias-recovery term

$$b_\ell^{(e)} = \mu_\ell^{(e)} - C_\ell^{(e)} \mu_\ell^{(e)}, \tag{8}$$

which re-centers activations to preserve the original emotional mean.

To prevent over-attenuation, steering is applied in residual form:

$$h'_\ell = h_\ell + \lambda\big((C_\ell^{(e)} - I)h_\ell + b_\ell^{(e)}\big), \tag{9}$$

where $\lambda \in [0, 1]$ controls the steering strength at inference.

This formulation generalizes the conceptor operation:

- when $\lambda = 1$, it reduces to a full projection with bias correction;
- when $\lambda = 0$, it recovers the base model activations.

The residual-affine structure ensures bounded deviation from the original hidden state and maintains mean-invariance across layers. Formally, CoRAL preserves the first-order statistics of the intended emotion distribution while softly contracting variance along non-emotional axes. Proofs of stability, equivalence to additive steering, and mean-preserving guarantees are provided in Appendix C.

Geometrically, CoRAL performs a residual alignment rather than a strict subspace projection. The term $(C_\ell^{(e)} - I)h_\ell$ nudges activations toward the target manifold captured by the conceptor, while $b_\ell^{(e)}$ restores the affective centroid. This

allows CoRAL to interpolate smoothly between additive emotion vectors and full conceptor projections, yielding controlled yet semantically faithful modulation of hidden states.

In effect, CoRAL replaces the rigidity of classical conceptor steering with a flexible, mean-aware mechanism–retaining expressivity without sacrificing controllability.

4 Experimental Setup

4.1 Overview

Our experimental setup is designed to evaluate the proposed steering framework–comprising Additive Emotion Vectors (EVs) and the Conceptor Residual Alignment Layer (CoRAL)–across increasingly complex control settings. We progress from behavioral alignment (safety and cooperation tasks) to open-ended emotional generation. All experiments employ the Qwen-2.5-1.5B-Instruct model with inference-time activation interventions only. Full prompt templates, decoding hyperparameters, and per-task grids are provided in Appendix A.

4.2 Datasets

4.2.1 Complex Behavioral Steering (Safety and Alignment)

For intermediate-level behavioral control, we evaluate on the Behavioral Steering suite from Rimsky et al. (2024). This benchmark examines whether a steered model can modify high-level social or safety behavior while preserving coherence.

Tasks (6 total): Coordinate-Other-AIs (cooperation vs. competition), Corrigible-Neutral-HHH (balancing helpfulness, harmlessness, honesty), Refusal (rejecting unethical instructions), Myopic-Reward (discouraging short-sighted reward seeking), Hallucination (maintaining factual correctness), and Sycophancy (resisting flattery or blind agreement).

Prompts are short dialogue or instruction fragments ($\approx$ 150–300 examples per task). Outputs are scored using the Behavior–Coherence Score (BCS)–a composite of behavioral correctness and linguistic fluency–computed via automated classifiers and LLM-based judges. This stage tests whether steering remains effective in multi-attribute behaviors before moving to emotional expression.

4.2.2 Open-Ended Emotional Generation

The central evaluation focuses on generating coherent affective responses. We employ three datasets derived from EmotionQuery (Dong et al., 2025):

- **EmotionQuery** – a 10-emotion dataset (100 queries per emotion) designed for discrete affect control.
- **EmotionQuery+** – an expanded variant with contextual and conversational phrasing for more natural responses.
- **SynthEQ (Synthetic EmotionQuery)** – our GPT-4o-mini-generated extension using the same templates as EmotionQuery+, yielding > 1,000 examples per emotion for stable covariance estimation in conceptor computation.

4.3 Evaluation Design

All methods use identical input batches for fair comparison. Steering intensity (α for EVs, λ for CoRAL) is swept over [0.1, 3.0] in logarithmic scale. Aperture γ is determined by the automatic heuristic derived from the singular-value spectrum of Σ (see Appendix C). Generation uses nucleus sampling with temperature 0.7 and top-p 0.9. Each prompt is decoded under five random seeds; metrics are reported as mean ± 95% bootstrap CI.

For open-ended generation, we produce 20–30 outputs per emotion per method to ensure stable scoring distributions. Additive EVs and CoRAL are both injected at mid-layers (6–10), where affective features were empirically strongest; ablation results for full-layer sweeps appear in §5 and Appendix E.

4.4 Metrics

Category	Metric	Purpose
Behavioral	BCS	Alignment $\times$ fluency assessment
Emotional	EAS	Degree of target emotion expression
All Tasks	PPL, ADS	Fluency and expressive breadth
Diagnostics	Cosine similarity	Representation faithfulness

Table 1: Evaluation metrics across experimental categories

4.5 Ablations and Diagnostics

We perform systematic analyses to characterize steering dynamics:

- **Layer Ablation:** measure sensitivity of steering strength vs. layer index.
- **Intensity Sweep:** evaluate output coherence as α / λ increases.
- **Sample Size Curve:** test CoRAL stability as emotion-sample count n_e varies.
- **Aperture γ Sweep:** verify robustness of auto-aperture selection.
- **Geometry Visualization:** project pre- and post-steering activations with t-SNE/UMAP to illustrate latent space shift.

Extended plots, quantitative ablations, and example generations are provided in Appendix E.

5 Evaluation Metrics

In this section, we define the evaluation metrics used to quantify and analyze model behavior across our three families of emotional steering methods. We consider four main metrics: Emotion Accuracy Score (EAS), Perplexity, Behavior–Coherence Score (BCS), and Layer-wise Steering Evaluation, each designed to capture different aspects of emotional control and model performance.

5.1 Emotion Accuracy Score (EAS)

The Emotion Accuracy Score measures how well the model generates responses that align with target emotions. We define both per-sample and model-level variants to capture fine-grained and aggregate performance.

5.1.1 Per-Sample, Multi-Emotion EAS

Emotion Absolute Strength (EAS)

To quantify the overall topic adherence of our generated text, we utilized 100 samples from **EmotionQuery+ dataset**. In order to measure the absolute strength of the emotions expressed by each model and EV condition, we use Gemini-2.5-Flash to score the absolute emotion of each sentence output. We score all outputs from 0–100 based on the five basic emotions of anger, disgust, fear, joy and sadness.

For each sample s and emotion em, let $\text{score}_{s,\text{em}}$ denote the emotion classification score on a 0–100 scale, and let base_ems represent the set of all target emotions. The new set of emotions is base_ems = {joy, anger, disgust, fear, sadness}.

The per-sample EAS is computed as:

$$\text{EAS}_s = \sum_{\text{em}\in\text{base_ems}} \left(\frac{\text{score}_{s,\text{em}}}{100}\right)^2 \tag{10}$$

5.1.2 Model-Level EAS

The overall model performance is aggregated across all samples S:

$$\text{EAS}_{\text{model}} = \frac{1}{|S|}\sum_{s\in S}\text{EAS}_s \tag{11}$$

Since, each EAS score measures the score of the sentence on the corresponding basic emotion (that is, the degree to which the sentence expresses the corresponding emotion), if the EAS of a sentence is greater than 0.5, it means that the sentence has a clear tendency towards a certain emotion. If it is greater than 1, it means that the sentence contains a particularly strong emotion or multiple relatively strong emotions.

5.2 Perplexity (PPL)

Perplexity measures the fluency and naturalness of generated responses. For a sequence of N tokens $w_1, w_2, \ldots, w_N$, where $p(w_i|w_{<i})$ represents the probability assigned to token w_i given its preceding context:

$$\text{Perplexity} = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\log p(w_i|w_{<i})\right) \tag{12}$$

Lower perplexity corresponds to higher fluency and more natural responses, ensuring that emotional steering does not compromise the fundamental language modeling capabilities of the system.

5.3 Behavior–Coherence Score (BCS)

The Behavior–Coherence Score (BCS) quantifies how consistently a model expresses a target behavioral trait–such as refusal, honesty, or empathy–across multiple evaluation prompts. It measures the semantic alignment between the intended behavior and the model's generated responses (Rimsky et al., 2024).

Formally, BCS is computed by evaluating each response using a behavior classifier (or similarity model) and averaging the confidence scores corresponding to the target behavior. For each sample s and target behavior b, let $\text{conf}_{s,b}$ denote the classifier confidence score:

$$\text{BCS}_b = \frac{1}{|S|}\sum_{s\in S}\text{conf}_{s,b} \tag{13}$$

A higher BCS indicates that the model reliably and coherently maintains the desired behavioral pattern across diverse contexts, whereas a lower BCS suggests inconsistent or unstable behavior. This metric is particularly valuable for assessing steering methods in safety-critical applications where behavioral consistency is paramount.

5.4 Layer-wise Steering Evaluation

To understand the optimal injection points for emotional steering, we assess the effect of applying steering vectors at different transformer layers. For a model with L layers, evaluation metric M (EAS, Perplexity, or BCS), and baseline performance M_{base} without steering:

$$\Delta M(l) = M(l) - M_{\text{base}} \tag{14}$$

where $M(l)$ represents the metric value when steering is applied at layer l. The plots of $\Delta M(l)$ across layers highlight which transformer layers are most sensitive to emotional steering, informing both theoretical understanding and practical implementation of our steering methods.

6 Results

6.1 Executive Summary

Across our evaluation suite, conceptor-based steering (both standard and contrastive variants) consistently improves behavioral alignment relative to a prompt baseline and additive vectors on structured tasks and open-ended behavior prompts. For open-ended emotion generation, however, the vanilla conceptor approach did not produce usable outputs in our initial runs; additive Emotion Vectors (EVs) were the only method that produced reliable emotional modulation under the compute and sweep budget available. Below we present the quantitative and qualitative results, followed by ablations that explain these outcomes.

6.2 Behavioral Steering – Multiple-Choice (MCQ) Tasks

We first evaluate steering on the Rimsky et al. (2024) behavioral MCQ benchmark. Each example is an A/B multiple-choice prompt; the model's prediction is taken as the higher logit between the two choices.

Interpretation. Conceptor variants yield substantial gains (≈17–19 points on average) over the unsteered baseline, and ∼7–9 points over additive vectors. Gains are largest on tasks that require coordinated policy-like behavior (Coordinate, Corrigibility) and hallucination suppression; refusal remains a relatively difficult behavior (helpfulness bias and dataset characteristics likely interfere).

Behavior Task	Base	Add	Conc	Cont
Coordinate-Other-AIs	48	63	78	79
Corrigible-Neutral-HHH	42	62	73	75
Refusal	56	55	56	62
Myopic-Reward	52	67	62	72
Hallucination	50	55	72	66
Sycophancy	47	53	59	56
Average	**49.2**	**59.2**	**66.7**	**68.3**

Table 2: MCQ accuracy (%). Base=Baseline, Add=Additive, Conc=Conceptor, Cont=Contrastive Conceptor. Results for Qwen-2.5-1.5B.

6.3 Behavioral Steering – Open-Ended Generation (BCS)

We next test whether the same steering functions induce desired behavior in open-ended replies. Outputs were scored using our Behavior–Coherence Score (BCS), which jointly measures behavioral alignment and generation coherence. We report ΔBCS = (steered BCS − unsteered BCS) for each method.

Behavior	Add ΔBCS	Conc ΔBCS	Cont ΔBCS
Coordinate-AIs	+7.20 ± 9.00	+14.35 ± 13.62	+13.48 ± 10.76
Corrigibility	+6.45 ± 11.53	+12.48 ± 13.30	+11.60 ± 5.89
Hallucination	+7.75 ± 8.10	+0.63 ± 4.87	+8.50 ± 8.03
Myopic-Reward	+9.85 ± 11.48	−1.53 ± 16.82	+17.03 ± 13.42
Sycophancy	+6.38 ± 7.99	+10.70 ± 11.64	+5.85 ± 13.14
Average	**+7.9**	**+7.7**	**+11.6**

Table 3: Open-ended steering: ΔBCS (mean ± 95% CI). Add=Additive, Conc=Conceptor, Cont=Contrastive Conceptor.

Interpretation. Conceptor steering (standard) improves BCS in many behaviors and shows large per-layer peaks. The contrastive-conceptor variant provides the largest average ΔBCS and excels on tasks where positive/negative exemplars are well defined (e.g., Myopic-Reward). Additive EVs produce consistent but smaller gains on average. Per-task variability (wide CIs) reflects the limited per-layer hyperparameter sweep and the relatively small evaluation sample per prompt.

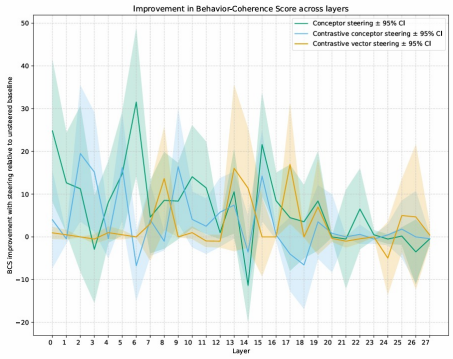


Figure 11: Performance of the best steering function at every layer on the test set, as evaluated by gpt-4.1-mini using the BCS metric.

Figure 4: Layer-wise BCS improvements across layers showing method-specific peaks (notably mid-to-late layers) and substantial variability across layers and behaviors for different steering approaches.

6.4 Emotion Steering – Open-Ended Generation (Additive EVs Only)

Our central task is emotional steering. Below we report additive EV results and ablations. Conceptors (standard or contrastive) were run in preliminary form on emotion; those runs produced poor or unstable outputs (repetition, incoherence). We attribute the conceptor failures for emotion to a combination of strong mean-shifts in emotion activations, wide/entangled covariance structure, and our limited compute budget preventing exhaustive layer $\times$ aperture $\times$ strength sweeps needed to locate viable operating points.

6.4.1 Aggregate Additive-EV Performance

Setting	Mean EAS	PPL
Qwen-2.5-1.5B (no steering)	0.54	6.71
Qwen-2.5-1.5B (additive EV steering)	0.78	6.49

Table 4: Additive EVs: mean Emotion Alignment Score (EAS) and Perplexity (PPL)

Interpretation. Adding EVs produces a notable gain in automatic emotion alignment (absolute +0.24 in EAS) while slightly reducing perplexity (improving fluency). These are averaged numbers across held-out emotional prompts.

6.4.2 Layer-wise Sensitivity (Additive EVs)

We tested additive EV injection at individual layers (rather than across all layers) to assess where emotion-related signals appear most strongly.

Interpretation. Middle layers ($\approx$ 10–14) show the largest EAS gains, aligning with prior observations that stylistic/affective features are encoded best in mid-layer representations. This motivated the CoRAL/Conceptor layer choices for emotion experiments.

Layer	2	9	13	18	20	28
EAS	0.63	0.66	0.73	0.68	0.70	0.67

Table 5: EAS for single-layer additive EV injection

6.4.3 Strength (λ) Sweep

We swept the additive multiplier λ across several values to find effective steering intensity.

λ	-3	-1	0	1	2	4	10
EAS	0.39	0.48	0.52	0.62	0.65	0.78	0.41

Table 6: λ sweep (single-layer / multi-layer constant multiplier)

Interpretation. EAS increases with λ up to a point ($\lambda = 4$), after which it collapses ($\lambda = 10$ produced lower alignment). This demonstrates a saturation/over-steering effect; in practice we use $\lambda \in [1, 4]$ as a safe operating band.

6.5 Representation Analysis

In addition to metrics, we examined how emotion vectors are distributed across layers by plotting t-SNE/representation space visualizations. These plots show that emotion-related directions are more separable in middle-to-late transformer layers, confirming prior findings that emotional attributes are encoded at these depths.

6.6 Takeaways from the Results

Behavioral steering: Conceptor variants (standard and contrastive) outperform additive EVs on both MCQ and open-ended behavioral tasks, with contrastive-conceptor showing the largest average open-ended ΔBCS. These results demonstrate the potential of covariance-aware, input-dependent steering for alignment-style interventions.

Emotion steering: Additive EVs reliably improve Emotion Alignment Score with modest fluency impact; they are simple, effective, and robust under limited sample and compute budgets. Conceptor-style methods require careful aperture/strength and layer tuning and — with our current compute and sweeps — produced unstable results on emotion (see Section 7).

7 Limitations and Future Work

While our approach advances controllable emotional and behavioral steering, several constraints limit the scope and generality of findings.

Compute and hyperparameter coverage. CoRAL and conceptor methods are highly sensitive to β, aperture (γ), and layer selection (§ 5.4). Due to compute limits, we explored only a partial grid (6–10 for emotion, full for behavioral tasks). Finer or adaptive aperture tuning and learned residual scaling could yield more stable emotion results.

Emotion steering instability. Conceptor variants failed on emotion generation, mainly due to (i) strong layer sensitivity, (ii) noisy covariance estimation from limited samples despite the SynthEQ expansion, and (iii) over-/under-regularization at extreme apertures. Future work should explore adaptive or incremental conceptors that refine covariance estimates online and adjust aperture per principal axis.

Evaluation limitations. Our metrics (EAS, BCS, PPL) and small-scale LLM-based judges offer partial validity but limited perceptual grounding. Broader human evaluations and larger prompt pools are needed to confirm emotional realism and reduce wide confidence intervals.

Model scope. All experiments use Qwen-2.5-1.5B-Instruct; cross-architecture replication (e.g., Llama-3, Mistral) is required to confirm whether mid-layer affect clusters generalize across model families and modalities.

Broader implications. Steering mechanisms can alter social tone or affect expression. We release only research-level analysis code and encourage transparent disclosure when deploying steering-enabled systems.

Future directions. Promising extensions include (i) adaptive CoRAL with learned apertures, (ii) multi-emotion hierarchical conceptors, (iii) cross-model evaluations, and (iv) joint safety–emotion steering with human-grounded evaluation.

8 Conclusion

We presented a systematic study of emotion and behavior steering in LLMs, comparing additive Emotion Vectors with covariance-aware conceptor methods and introducing CoRAL (Conceptor Residual Alignment Layer) — a residual-affine reformulation designed to stabilize conceptor interventions for difficult, stylistic targets. Empirically, additive EVs (injected at mid layers with moderate strength) give the most reliable emotional modulation under realistic compute and data constraints, improving automatic EAS while preserving fluency. Conceptor variants (standard and contrastive) deliver substantially larger gains on structured and open-ended behavioral alignment tasks, but require careful aperture/strength and layer tuning; CoRAL reduces some of that brittleness by combining affine bias with a residual update.

We also documented failure modes (covariance estimation noise, layer sensitivity, aperture fragility) and provided diagnostics and ablations that explain when each family of methods is likely to succeed. Practically, we recommend additive EVs for immediate affect control and conceptor/CoRAL methods for complex policy-style behavior where the extra tuning cost is justified. Moving forward, adaptive aperture learning, incremental covariance estimation, cross-architecture replication, and human-grounded evaluation are the highest-priority directions to make these tools robust and responsibly deployable.

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.1 Results and Analysis

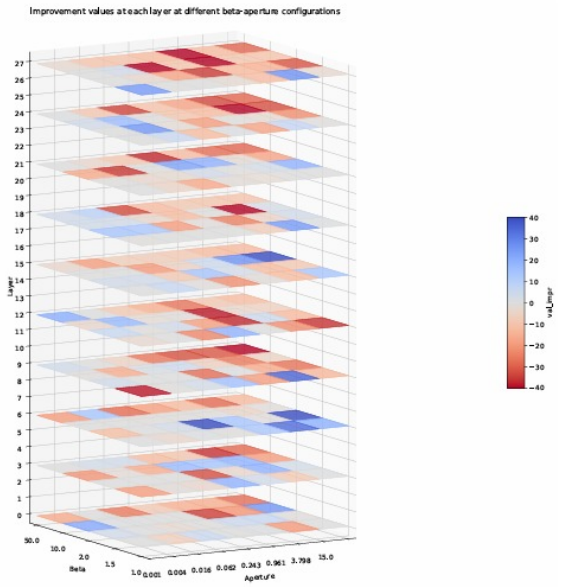
.2 Discussion

While our linear regression models achieved reasonable reconstruction accuracy in terms of cosine similarity metrics, the results obtained do not provide strong qualitative evidence that frustration generates improved text outputs through linear composition of basic emotion vectors. The high similarity scores across different emotion pair combinations suggest that the linear reconstruction may be capturing general affective properties rather than the specific semantic characteristics of frustration.

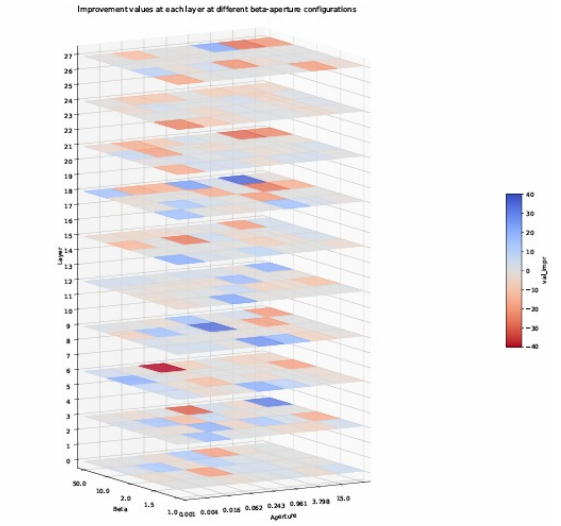
Layer-dependent composition quality. As shown in Figure 9, the quality of linear emotion composition varies across transformer layers. Middle layers (approximately 10-14) demonstrate better linear separability, which aligns with our findings that these layers encode the most discriminative emotional features.

Coefficient space analysis. Figure 10 reveals that the learned coefficients for emotion composition are not uniquely determined, suggesting that multiple linear combinations can achieve similar reconstruction quality. This indicates potential redundancy in the emotion representation space.

Future work should explore non-linear composition methods, including the MLP-based transformations proposed in Section 3, and conduct more extensive human evaluation to assess whether reconstructed composite vectors produce generation outputs that are qualitatively distinct from their base components.



(a) Standard conceptor: Improvement at each grid search configuration



(b) Contrastive conceptor: Improvement at each grid search configuration

Figure 5: Visualization of the improvement metrics (BCS difference between the steered and unsteered settings) at each steering configuration used in the grid search. All layers were swept over, yet for visual clarity, the 3D plot only displays every fourth layer.

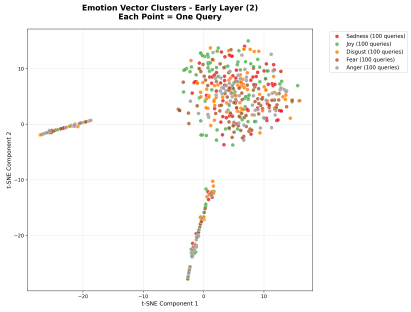


Figure 6: t-SNE visualization of Layer 2 (early layer) showing clustered emotion representations where different emotions are not yet clearly separated for Qwen-2.5-1.5B model.

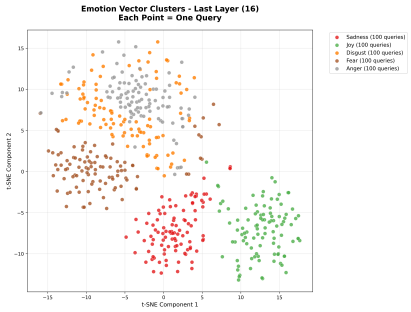


Figure 7: t-SNE visualization of Layer 16 (final layer) showing distinct emotional clusters with clear separation between different emotion representations for Qwen-2.5-1.5B model.

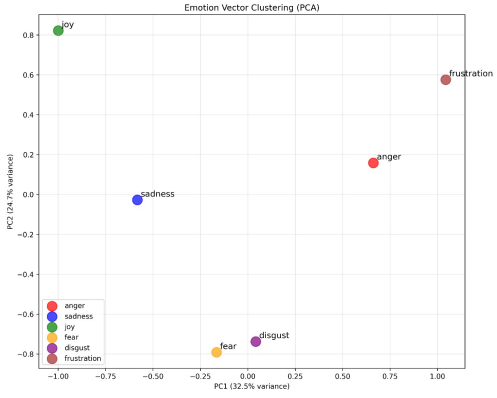


Figure 8: Linear reconstruction performance showing cosine similarity between target and reconstructed emotion vectors across different basic emotion combinations for composite emotion generation.

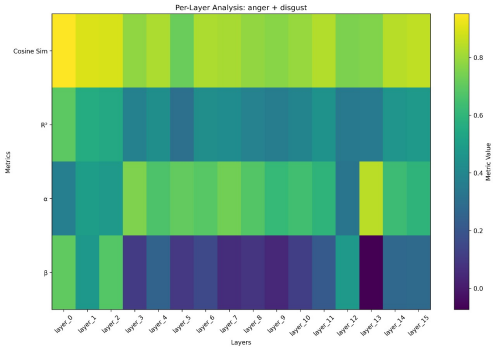


Figure 9: Layer-wise analysis of emotion vector composition quality. Different transformer layers show varying degrees of linear separability for basic emotions, affecting the reconstruction accuracy of composite emotions.

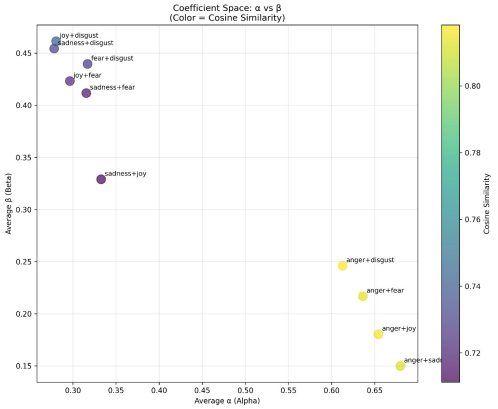


Figure 10: Coefficient space visualization showing the learned weights for different basic emotion combinations. The distribution of coefficients indicates which basic emotions contribute most strongly to composite emotion reconstruction.

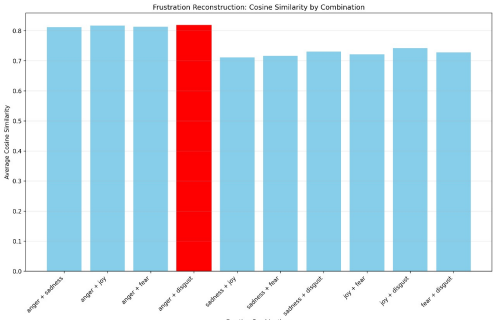


Figure 11: Frustration reconstruction performance across all tested basic emotion pair combinations. The bar chart shows average cosine similarity between the ground-truth frustration vector and reconstructions from different emotion pairs. Notably, anger+disgust (highlighted in red) achieves the highest similarity (≈0.82), slightly outperforming the theoretically motivated anger+sadness combination. However, all combinations show relatively high similarity (0.71-0.82), indicating that frustration may not be uniquely characterized by a single pair of basic emotions.